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Submanifolds

1.1 Embedded Submanifolds

1.1.1 Slice Charts for Embedded Submanifolds

Definition 1.1.1

If U is an open subset of $\mathbb{R}^n$ and $k \in \{0, \dots, n\}$, a **k -dimensional slice of U** (or simply a **k -slice**) is any subset of the form

$$S = \{(x^1, \dots, x^k, x^{k+1}, \dots, x^n) \in U : x^{k+1} = c^{k+1}, \dots, x^n = c^n\}$$

for some constants $c^{k+1}, \dots, c^n$.

Definition 1.1.2 ► local k -slice condition

Let M be a smooth n -manifold.

- Let (U, φ) be a smooth chart on M . If S is a subset of U such that $\varphi(S)$ is a k -slice of $\varphi(U)$, then we say that S is a **k -slice of U** .
- Given a subset $S \subseteq M$ and a nonnegative integer k , we say that S satisfies the **local k -slice condition** if each point of S is contained in the domain of a smooth chart (U, φ) for M such that $S \cap U$ is a single k -slice in U .
- Any such chart is called a **slice chart for S in M** , and the corresponding coordinates $(x^1, \dots, x^n)$ are called **slice coordinates**.

1.1.2 Level Sets

Definition 1.1.3

If $\Phi : M \rightarrow N$ is a smooth map.

- A point $p \in M$ is said to be a **regular point of Φ** if $d\Phi_p : T_p M \rightarrow T_{\Phi(p)} N$ is surjective; it is a **critical point of Φ** otherwise.
- A point $c \in N$ is said to be a **regular value of Φ** if every point of the level set $\Phi^{-1}(c)$ is a regular point, and a **critical value** otherwise.

Theorem 1.1.4 ▶ Constant-Rank Level Set Theorem

Let M and N be smooth manifolds, and let $\Phi : M \rightarrow N$ be a smooth map with constant rank r . Each level set of Φ is a properly embedded submanifold of codimension r in M .

Proof. Write $m = \dim M$, $n = \dim N$, and $k = m - r$. Let $c \in N$ be arbitrary, let S denote the level set $\Phi^{-1}(c) \subseteq M$. From the rank theorem, for each $p \in S$, there are smooth charts (U, φ) centered at p and (V, ψ) centered at $c = \Phi(p)$ in which Φ has a coordinate representation of the form

$$\psi \circ \Phi \circ \varphi^{-1}(x^1, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0)$$

and therefore $S \cap U$, then

$$\begin{aligned} q \in S &\iff \Phi(q) = c \iff (x^1(q), \dots, x^r(q), 0, \dots, 0) = (0, \dots, 0) \\ &\iff x^1(q) = \dots = x^r(q) = 0 \end{aligned}$$

Thus

$$S \cap U = \{(x^1, \dots, x^r, x^{r+1}, \dots, x^m) \in U : x^1 = \dots = x^r = 0\}$$

is the k -slice. So S satisfies the local k -slice condition, so it is an embedded submanifold of dimension k . $\square$

1.2 Restricting Maps to Submanifolds

Theorem 1.2.1 ▶ Restricting the Domain of a Smooth Map

If M and N are smooth manifolds with or without boundary, $F : M \rightarrow N$ is a smooth map, and $S \subseteq M$ is an immersed or embedded submanifold, then $F|_S : S \rightarrow N$ is smooth.

Theorem 1.2.2 ▶ Restricting the Codomain of a Smooth Map

Suppose M is a smooth manifold (without boundary), $S \subseteq M$ is an immersed submanifold, and $F : N \rightarrow M$ is a smooth map whose image is contained in S . If F is continuous as a map from N to S , then $F : N \rightarrow S$ is smooth.

1.3 The Tangent Space to a Submanifold

Definition 1.3.1

Let M be a smooth manifold with or without boundary, and let $S \subseteq M$ be an **immersed** or embedded submanifold. Since the inclusion map $\iota : S \hookrightarrow M$ is a smooth immersion, at each point $p \in S$ we have an injective linear map $d\iota_p : T_p S \rightarrow T_p M$. In terms of derivations, this injection works in the following way: for any vector $v \in T_p S$, the image vector $\tilde{v} = d\iota_p(v) \in T_p M$ acts on smooth functions on M by

$$\tilde{v}f = d\iota_p(v)f = v(f \circ \iota) = v(f|_S).$$

Remark.

We adopt the convention of **identifying** $T_p S$ with its image under this map, thereby thinking of $T_p S$ as a certain linear subspace of $T_p M$

Proposition 1.3.2

Suppose M is a smooth manifold with or without boundary, $S \subseteq M$ is an immersed or embedded submanifold, and $p \in S$. A vector $v \in T_p M$ is in $T_p S$ if and only if there is a smooth curve $\gamma : J \rightarrow M$ whose image is contained in S , and which is also smooth as a map into S , such that $0 \in J$, $\gamma(0) = p$, and $\gamma'(0) = v$.

Proof. $\Leftarrow$: If there is $\gamma : J \rightarrow M$ satisfying

1. $\text{Im}(\gamma) \subseteq S$
2. $\tilde{\gamma} : J \rightarrow S$ is smooth, where $\iota \circ \tilde{\gamma} = \gamma$
3. $\gamma(0) = p, \gamma'(0) = v$.

Then since

$$v = \gamma'(0) = (\iota \circ \tilde{\gamma})'(0) = d\iota_p(\tilde{\gamma}'(0))$$

$\tilde{\gamma}$ is a smooth curve on S , $\tilde{\gamma}'(0) \in T_p S$, thus $v \in T_p S \subseteq T_p M$.

$\Rightarrow$ if $v \in T_p S \subseteq T_p M$, then there exists $u \in T_p S$ such that $d\iota_p(u) = v$. There exists a smooth curve $\tilde{\gamma} : J \rightarrow S$ such that $\tilde{\gamma}(0) = p$, $\tilde{\gamma}'(0) = u$. Let $\gamma = \iota \circ \tilde{\gamma}$, then

$$\text{Im } \gamma \subseteq S, \quad \gamma(0) = \iota(\tilde{\gamma}(0)) = \iota(p) = p, \quad \gamma'(0) = d\iota_p(\tilde{\gamma}'(0)) = d\iota_p(u) = v$$

□

Proposition 1.3.3

Suppose M is a smooth manifold, $S \subseteq M$ is an **embedded** submanifold, and $p \in S$. As a subspace of $T_p M$, the tangent space $T_p S$ is characterized by

$$T_p S = \{v \in T_p M : vf = 0 \text{ whenever } f \in C^\infty(M) \text{ and } f|_S = 0\}.$$

Proof. First we assume that $v \in T_p S \subseteq T_p M$. This means, more precisely, there exists $w \in T_p S$ such that $d\iota_p(w) = v$. Then for each $f \in C^\infty(M)$ with $f|_S = 0$, we have

$$vf = d\iota_p(w)f = w(f \circ \iota) = 0$$

Conversely, if $v \in T_p M$ satisfies $vf = 0$ whenever $f \in C^\infty(M)$ and $f|_S = 0$. Let $(x^1, \dots, x^n)$ be slice coordinates for S in some neighborhood U of p , so that $U \cap S$ is the subset of U where $x^{k+1} = \dots = x^n = 0$, and $(x^1, \dots, x^k)$ are coordinates for $S \cap U$. Then $\iota : S \cap U \hookrightarrow M$ is represented

$$\iota(x^1, \dots, x^k) = (x^1, \dots, x^k, 0, \dots, 0)$$

Write

$$v = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i} \Big|_p,$$

Let $\varphi \in C^\infty(M)$ be a bump supported in U and is equal to 1 in a neighborhood of p . Take any $j > k$, let extend $f = \varphi(x) x^j$ to be zero on $M \setminus \text{supp } \varphi$. Then f is identically zero on S , we have

$$0 = vf = \varphi(p) v^j = v^j$$

□

Proposition 1.3.4

Suppose M is a smooth manifold and $S \subseteq M$ is an embedded submanifold. If $\Phi : U \rightarrow N$ is any local defining map for S , then $T_p S = \ker d\Phi_p : T_p M \rightarrow T_{\Phi(p)} N$ for each $p \in S \cap U$.

Proof. Let $\iota : S \hookrightarrow M$, then $\Phi \circ \iota$ is identically zero on $U \cap S$. Then $d\Phi_p \circ d\iota_p = 0$,

thus $\text{Im } d\iota_p \subseteq \ker d\Phi_p$. The other side, since $d\Phi_p$ is surjective, we have

$$\dim \ker d\Phi_p = \dim T_p M - \dim T_{\Phi(p)} N = \dim T_p S = \dim \text{Im } d\iota_p$$

Thus $\text{Im } d\iota_p = \ker d\Phi_p$. Since $\text{Im } d\iota_p$ is identified with $T_p S$, we have $T_p S = \ker d\Phi_p$. $\square$